KSJ RESEARCH

Research Note Vol.1

# Is AI Power Demand Real, and Does the Market Already Know?

Demand, ground truth, earnings and market pricing from official data — and what it means for IPO pricing

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# Demand is real, it shows up on the ground — earnings led the market, but P/E also rose from 13.5x to 40.1x

**IEA 2030E 945 TWh (2.3x 2024) · Virginia commercial sales +73% vs U.S. +10% · 4 names' market cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)**

## Key metrics [*1,*2,*3,*4,*5]

| Key metric | Value | Page |
|---|---|---|
| 2030E data-centre power | **945 TWh (2.3x)** | 1 |
| 2035E outlook range | 700~1,700 TWh | 1 |
| 2025 growth: AI DC · DC · world | +50% · +17% · +3% | 1 |
| Virginia commercial index (2025) | **173 vs U.S. 110** | 2 |
| 4 names' combined cap (2023-01→2026-10) | **7.7x** | 4 |
| Simple average / range | 14.6x / 4.4x~26.1x | 4 |
| 3-name earnings · multiple share | **61% · 39%** | 4 |
| 3-name P/E (start-2023→latest) | **13.5 → 40.1x** | 4 |
| KTB 10y (start-2023→latest) | 3.81 → 4.21% | 5 |

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 IEA, Key Questions on Energy and AI (Apr 2026) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 Korea Exchange (KRX) Open API, KOSPI daily trading *5 FSS DART annual reports (consolidated)
Note: KSJ Research calculations; see footnotes on each page. Judgements are the author's own

**2.3x**

Data-centre power 2030E ÷ 2024 (IEA Base Case)

### Demand is real — 945 TWh by 2030; AI data-centre demand +50%

- 2025: data centres +17%, AI data centres +50% vs world +3%
- Grew 12%/yr in 2017–2024; 1.5% of world power in 2024; U.S. 45%
- But the 2035 range is wide: 700–1,700 TWh (±42% vs base)

**+73%**

Virginia commercial power sales 2015→2025 (EIA)

### It shows up on the ground — 63 index points ahead of the U.S. (+10%)

- CAGR Virginia 5.6% vs U.S. 0.9% (2015–2025)
- Last 3 years (2022→2025) CAGR 6.8% vs 2.4%
- U.S. commercial price +26% since 2015 (volume +9.8%)

**7.7x**

4 names, combined cap, 2023-01-02→2026-10-19 (range 4.4x–26.1x)

### Earnings led — 61% of the gain for 3 names (ex. Doosan Enerbility) is earnings growth

- Simple average 14.6x — Doosan Enerbility (4.4x), 72% of starting cap, drags the total
- 3-name aggregate P/E 13.5x → 40.1x (start-2023 ↔ FY2022A, latest ↔ FY2025A)
- Hyosung Heavy Industries has the largest earnings share (88%) — only LS ELECTRIC was multiple-led (57%)

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# Contents

**Demand → ground → earnings → market → implications: the conclusion of each section**


I **Demand** 1

Data-centre power use reaches 2.3x by 2030; AI data-centre demand +50% in 2025

II **Ground truth** 2

Virginia commercial sales +73% since 2015, 63 index points ahead of the U.S.

III **Earnings** 3

NI to parent of 3 names 5.5x explains 61% of the market-cap gain

IV **Market** 4

For 3 names, 61% of the gain is earnings growth, 39% multiple expansion

V **Implications** 5

An earnings-led sector, yet 40x P/E — hence Vol.2 data-centre PF, Vol.3 IPO pricing and overseas peers

App. **Appendix** A1–A4

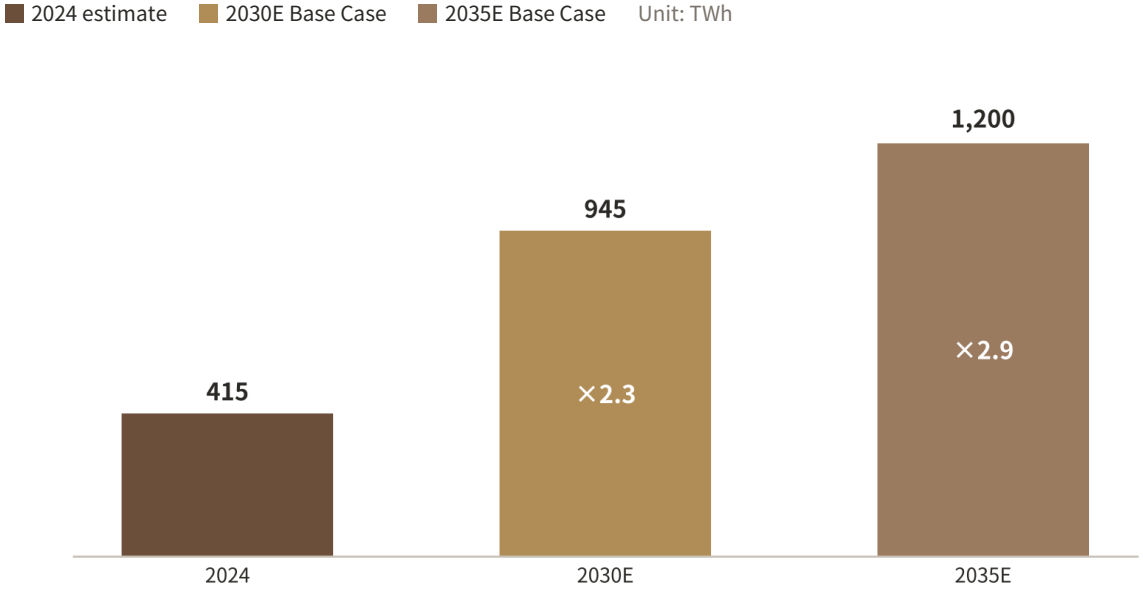
A1 Q&A · A2 Data notes (sources, datasets, dates, share-count check) · A3 Methodology · A4 Glossary


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# Demand is real: data-centre power use reaches 2.3x by 2030; AI data-centre demand +50% in 2025

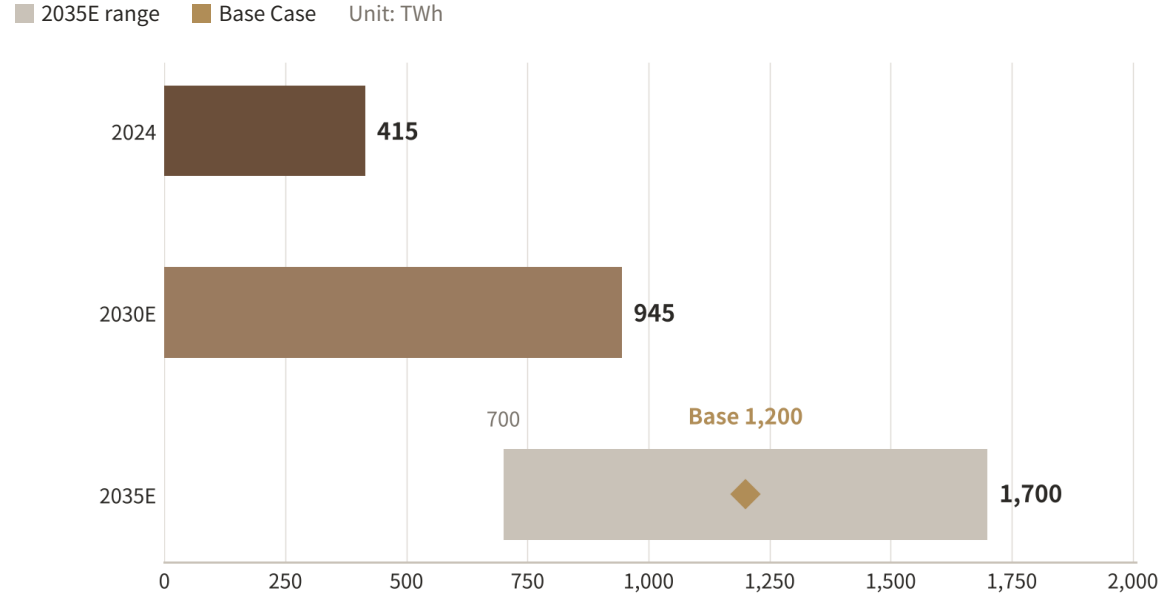
## IEA Base Case: 415 TWh (2024) → 945 TWh (2030) → 1,200 TWh (2035) — but the 2035 range (700–1,700 TWh) is wide

### Data-centre power use 2030E 945 TWh — 2.3x 2024 *1



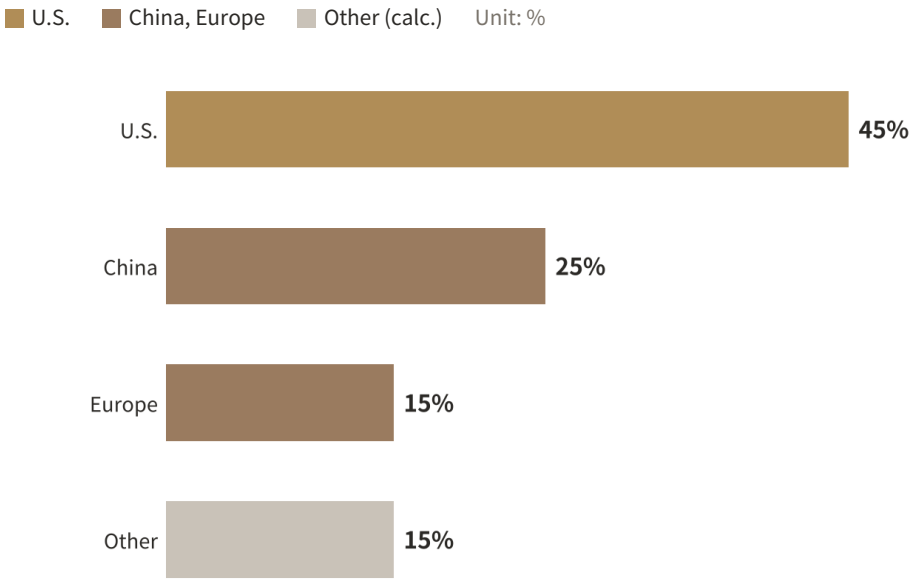

Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: KSJ Research. Multiples inside bars = year ÷ 2024

### 2035 range 700–1,700 TWh — −42% to +42% vs Base Case *1



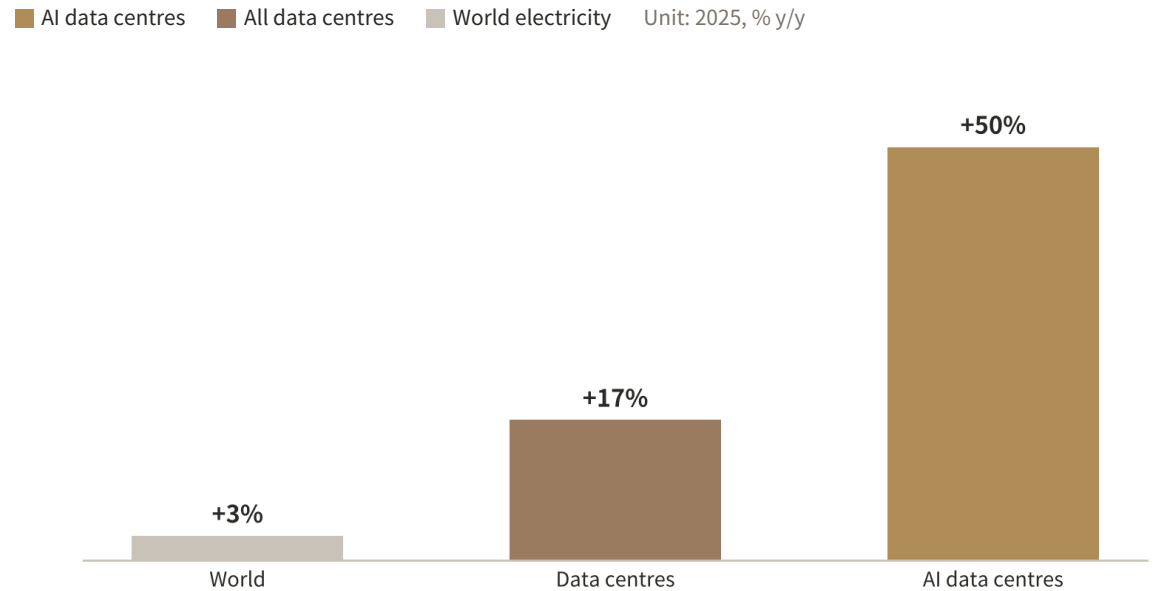

Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Range = IEA's 2035 low–high; % vs Base Case (KSJ Research)

### The U.S. used 45% of data-centre power in 2024; U.S., China and Europe 85% *1




Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Other = 100 − U.S., China, Europe (KSJ Research). Data-centre power grew 12%/yr in 2017–2024; 1.5% of world power in 2024

### AI data-centre demand +50% in 2025 — 17x the pace of world power (+3%) *1



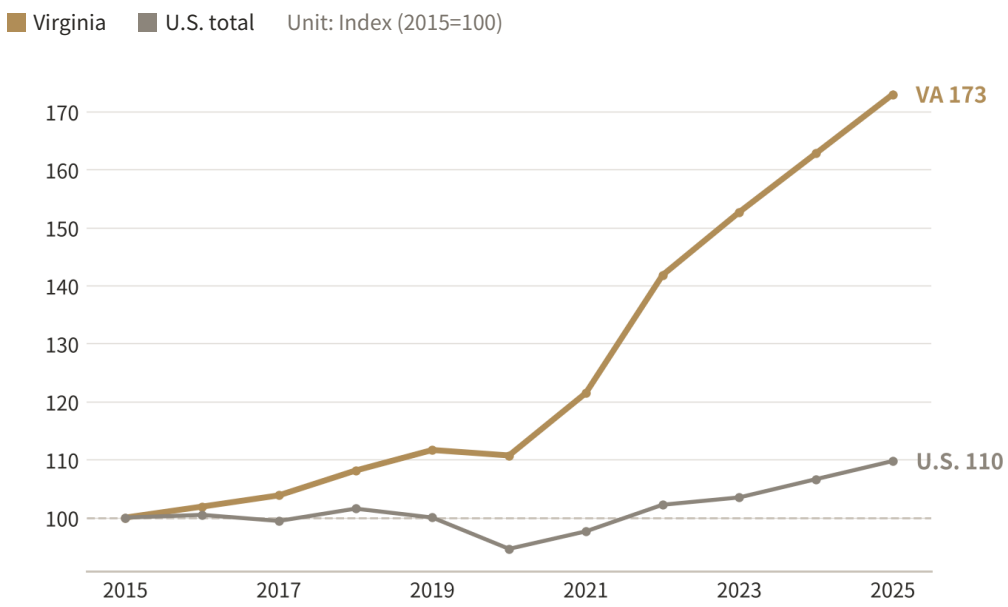

Source: *1 IEA, Key Questions on Energy and AI (Apr 2026)
Note: Multiple = 50 ÷ 3 (KSJ Research)

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# It shows up on the ground: Virginia commercial sales +73% since 2015, 63 index points ahead of the U.S. (+10%)

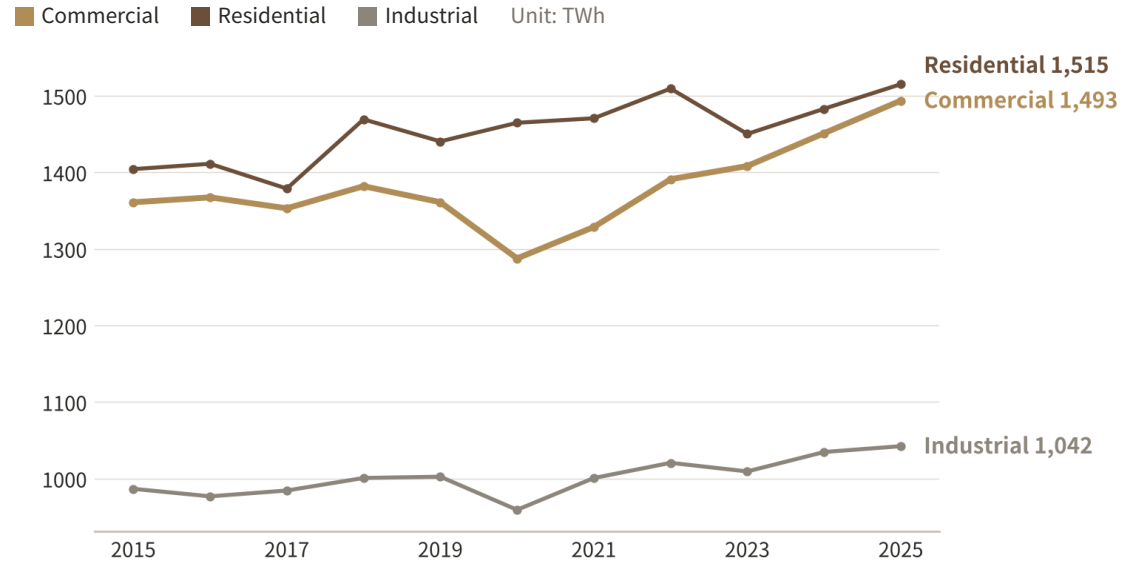
## Virginia, known as a data-centre hub, grew commercial sales 4.7pp a year faster than the U.S. — U.S. commercial prices rose +26%

### Virginia commercial sales +73% vs U.S. +10% — 63-index-point gap in 2025 *1



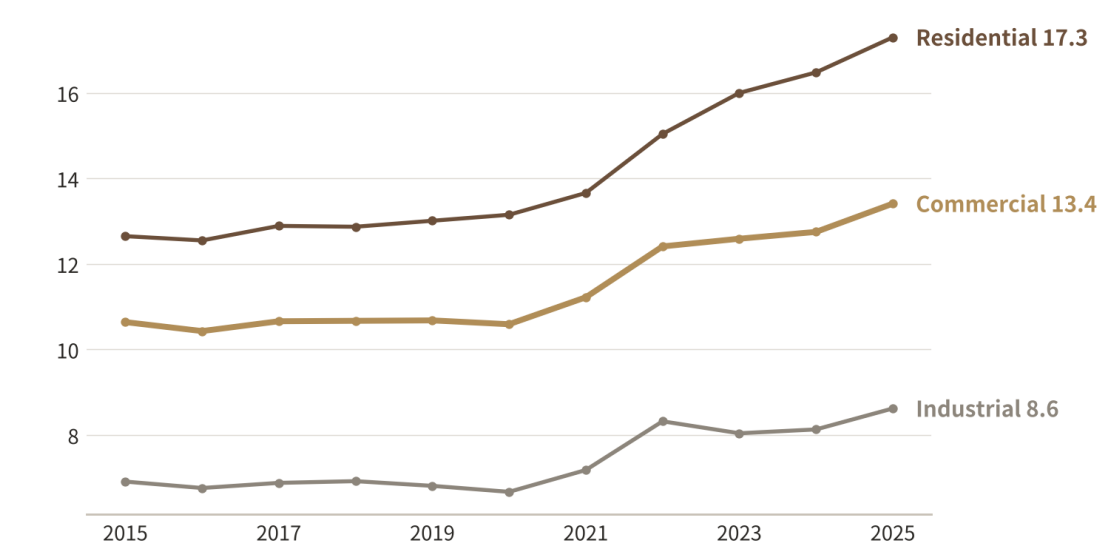

Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Index = sales ÷ 2015 × 100 (KSJ Research). CAGR Virginia 5.6% vs U.S. 0.9%

### U.S. sales by sector 2015→2025: Commercial +9.8% · Residential +7.9% · Industrial +5.6% *1




Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Sales in million kWh ÷ 1,000 = TWh (KSJ Research). Includes the 2020 commercial dip

### Commercial average price 10.64¢ → 13.41¢ (+26%) — faster rise since 2020 *1




Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial price CAGR 2015–2020 -0.1% → 2020–2025 4.8% (KSJ Research)

### Last 3 years CAGR: Virginia 6.8% vs U.S. 2.4% *1

Unit: % y/y; gap in pp

| Year | Virginia | U.S. total | Gap |
|---|---|---|---|
| 2018 | +4.1% | +2.1% | +2.0%p |
| 2019 | +3.3% | -1.5% | +4.8%p |
| 2020 | -0.8% | -5.4% | +4.6%p |
| 2021 | +9.7% | +3.2% | +6.5%p |
| 2022 | +16.7% | +4.7% | **+12.0%p** |
| 2023 | **+7.7%** | +1.2% | +6.4%p |
| 2024 | **+6.6%** | +3.0% | +3.6%p |
| 2025 | **+6.2%** | +2.9% | +3.3%p |

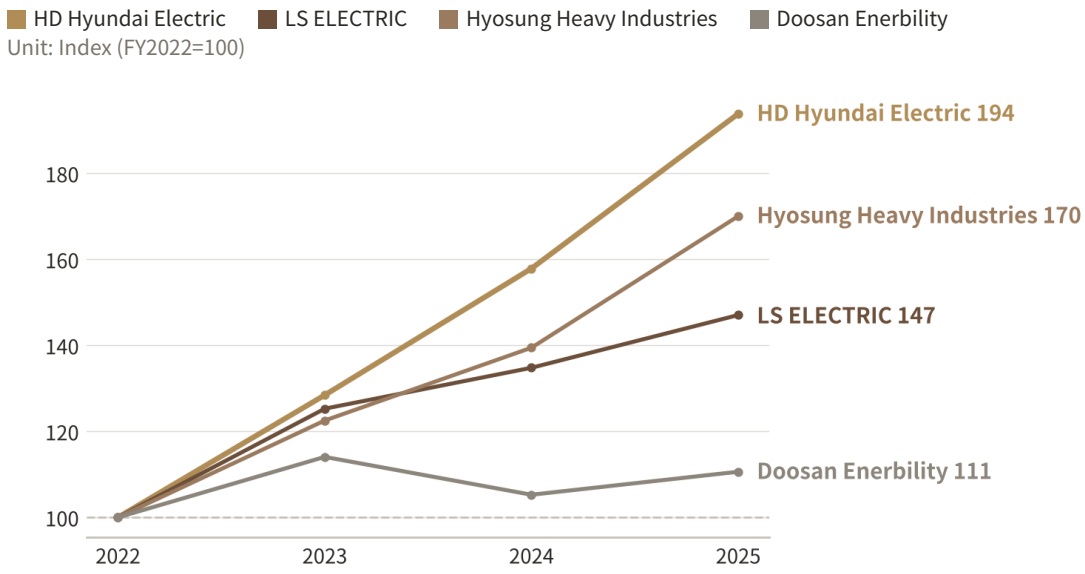
Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial sales growth y/y (KSJ Research). Shaded = last 3 years; gold = largest gap (2022)

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# Earnings grew too: NI to parent of 3 names (Doosan Enerbility excluded: FY2022 net loss) 5.5x explains 61% of the market-cap gain (16.2x)

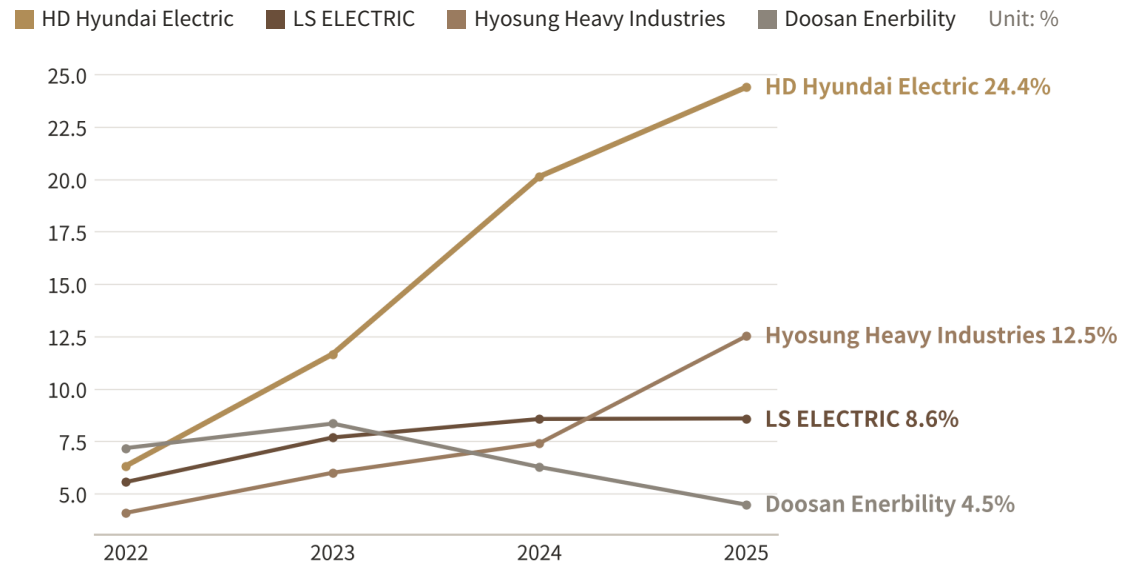
## Combined revenue of 4 names +31%, operating profit of 4 names +87% (FY2022A→FY2025A, consolidated)

### Revenue index (FY2022=100): HD Hyundai Electric highest at 194 · Doosan Enerbility 111 — combined +31% *1



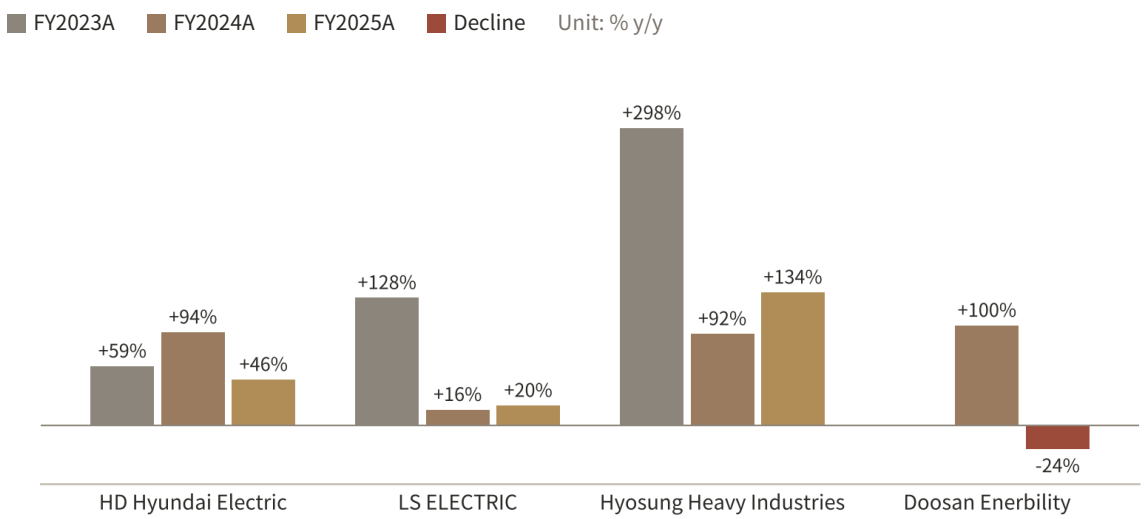

Source: *1 FSS DART annual reports (consolidated)
Note: Index = revenue ÷ FY2022A × 100 (KSJ Research). Consolidated; absolute values in the FY2025A table.

### Operating margin FY2025A: HD Hyundai Electric highest at 24.4%; 3 of 4 improved *1




Source: *1 FSS DART annual reports (consolidated)
Note: Operating margin = operating profit ÷ revenue (KSJ Research).

### NI to parent FY2022A→FY2025A: 3 names (Doosan Enerbility excluded: FY2022 net loss) combined 5.5x; Hyosung Heavy Industries leads at 17.8x; Doosan Enerbility turned profitable *1




Source: *1 FSS DART annual reports (consolidated)
Note: NI attributable to owners of the parent; no growth rate after a loss year (no bar). Combined multiple Doosan Enerbility excluded: FY2022 net loss.

### FY2025A combined: revenue 320,717, OP 29,314, NI 16,239 (KRW 100mn) *1

Unit: KRW 100mn, %, x

| Company | Revenue | OP | OPM | NI FY2022 | NI FY2025 | NI x |
|---|---|---|---|---|---|---|
| HD Hyundai Electric | 40,795 | 9,953 | 24.4% | 1,625 | 7,326 | 4.5x |
| LS ELECTRIC | 49,658 | 4,264 | 8.6% | 903 | 2,866 | 3.2x |
| Hyosung Heavy Industries | 59,685 | 7,470 | 12.5% | 291 | 5,199 | **17.8x** |
| Doosan Enerbility | 170,579 | 7,627 | 4.5% | -7,725 | 848 | Turned profitable |
| **Total** | **320,717** | **29,314** | **9.1%** | **-4,906** | **16,239** | **5.5x (3 names)** |

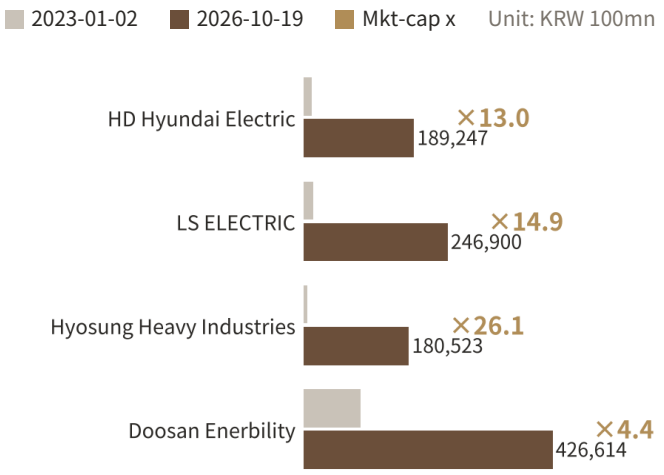
Source: *1 FSS DART annual reports (consolidated)
Note: Revenue, OP and OPM are FY2025A consolidated. NI x = FY2025A ÷ FY2022A; a label replaces the multiple when the base is ≤0. Combined multiple Doosan Enerbility excluded: FY2022 net loss (KSJ Research)

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# Earnings led: for 3 names (ex. Doosan Enerbility), 61% of the gain is earnings growth, 39% multiple expansion

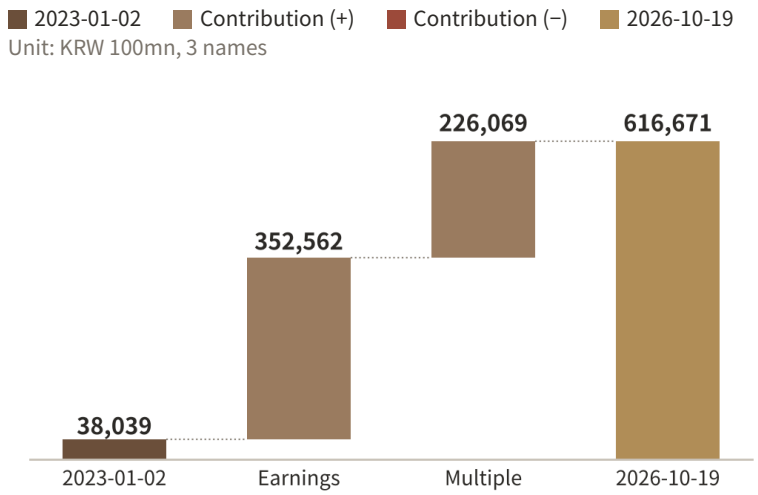
**3-name aggregate P/E 13.5x → 40.1x · Doosan Enerbility (72% of starting cap) is excluded from the split: FY2022A net loss — separate from the 4-name combined 7.7x**

## Market cap: 7.7x combined · 14.6x simple average — 4.4x to 26.1x by name *1,*2



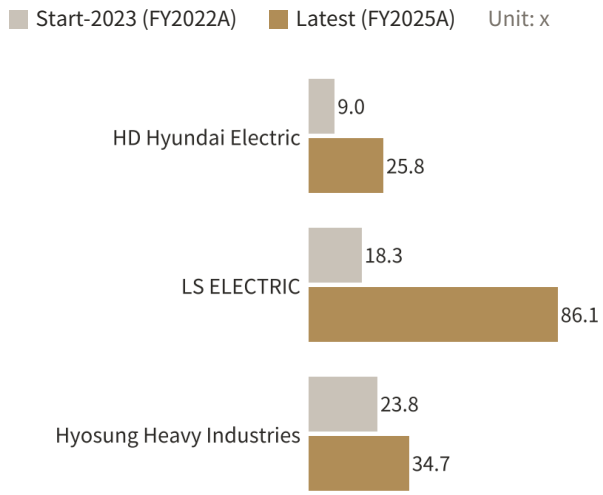

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Market cap = close × listed shares at each date (KRX MKTCAP). The combined multiple is below the average because Doosan Enerbility (4.4x) was 72% of the starting cap. LS ELECTRIC: 5:1 split on 2026-04-13 — we use the market-cap multiple 14.9x, not the raw price multiple 3.0x (split-adjusted 14.9x).

## Cap gain for 3 names (ex. Doosan Enerbility): 578,632 = earnings 352,562 + multiple 226,069 (KRW 100mn) *1,*2



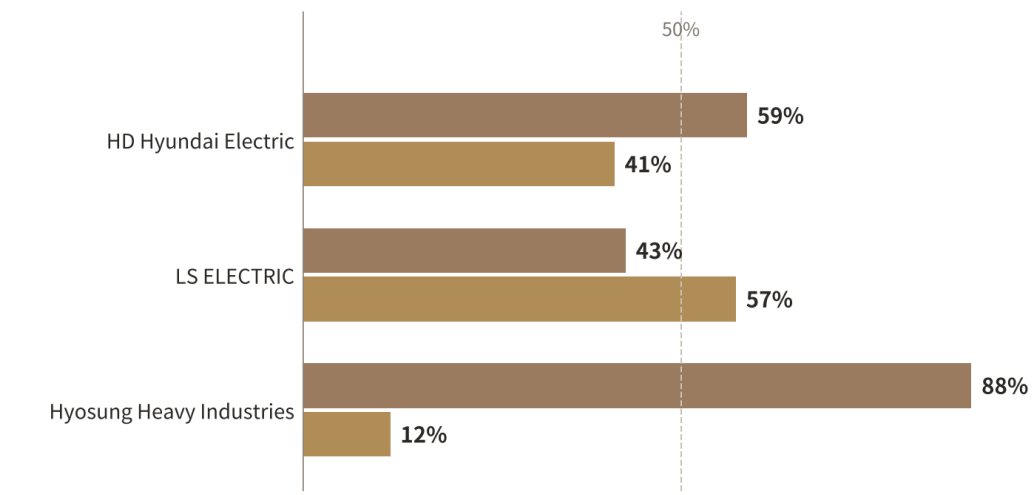

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Scope: 3 names (Doosan Enerbility excluded: FY2022A net loss). KRW split = Δcap × each term's log share. Log split Δln(cap) = Δln(NI) + Δln(P/E). Start-2023 cap ↔ FY2022A, latest cap ↔ FY2025A NI to parent. Only names with positive NI in both years are split (KSJ Research)

## 3-name P/E 13.5x → 40.1x; LS ELECTRIC highest at 86.1x *1,*2




Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: P/E = cap ÷ NI to parent. Doosan Enerbility: not shown, no starting P/E (FY2022A net loss). Including Doosan Enerbility, 4-name aggregate 64.2x (Doosan Enerbility 503x: denominator distortion from FY2025A NI of 848, KRW 100mn)

## Hyosung Heavy Industries has the largest earnings share (88%) — only LS ELECTRIC was multiple-led (57%) *1,*2




Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Earnings share = Δln(NI)/Δln(cap); multiple share = Δln(P/E)/Δln(cap); they sum to 100%. Dashed = 50%. Doosan Enerbility: not split (FY2022A net loss)

## 3-name aggregate P/E 13.5x → 40.1x: NI 5.5x × P/E 3.0x = cap 16.2x *1,*2

Unit: x; P/E start-2023 → latest; shares in %

| Company | Cap x | NI x | P/E (start → latest) | Earnings | Multiple |
|---|---|---|---|---|---|
| HD Hyundai Electric | 13.0x | 4.5x | 9.0 → 25.8 | 59% | 41% |
| LS ELECTRIC | 14.9x | 3.2x | 18.3 → 86.1 | 43% | **57%** |
| Hyosung Heavy Industries | 26.1x | 17.8x | 23.8 → 34.7 | 88% | 12% |
| Doosan Enerbility | 4.4x | Turned profitable | — | Not split (FY2022A net loss) | |
| **3-name total** | **16.2x** | **5.5x** | **13.5 → 40.1** | **61%** | **39%** |

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: P/E = cap ÷ NI (start-2023 ↔ FY2022A, latest ↔ FY2025A). Cap x = NI x × P/E x. Shaded = multiple-led. A label replaces the multiple when the base NI is ≤0

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# An earnings-led sector, yet 40x P/E — hence Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers

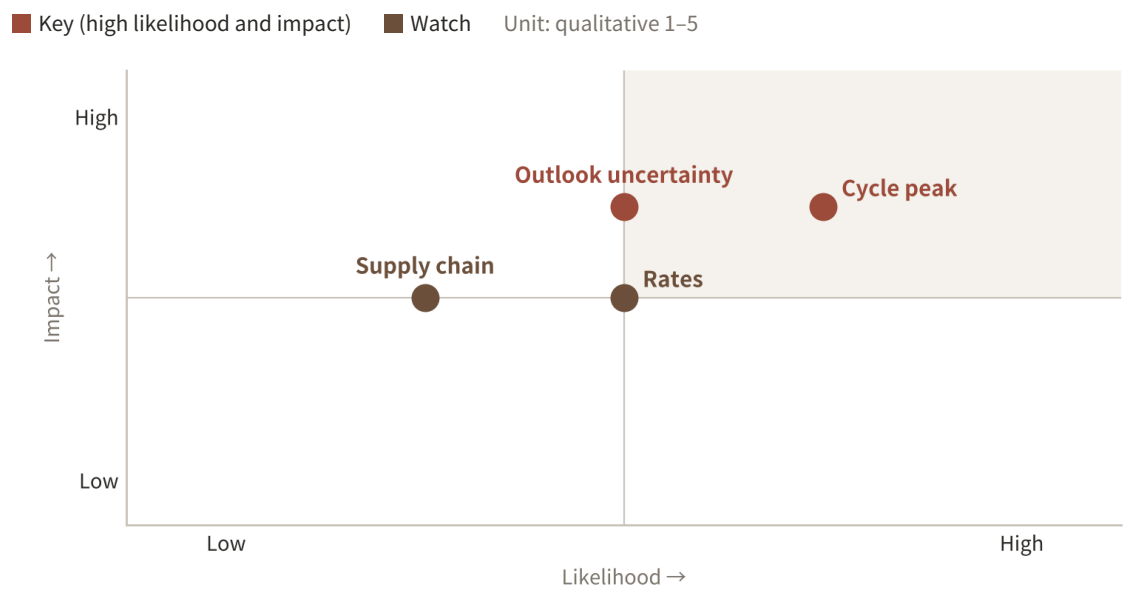
## Aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x · three lenses (ECM pricing · data-centre PF · capacity) and four risks

### Three lenses: data → judgement [*1,*2,*3,*4]

| Lens | What the data says | Implication |
|---|---|---|
| ECM pricing | aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x; cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x) | Earnings explain most of the gain, but P/E also rose 13.5x→40.1x. Rather than applying peer P/E as is, test earnings durability and apply an IPO discount |
| Data-centre PF | Virginia commercial sales 5.6%/yr; U.S. commercial price +26% (2015→2025) | Where volume and price rise together, PPAs and grid-connection timing are the key PF cash-flow variables |
| Capacity options | Revenue +31% (FY2022A→FY2025A); top OPM 24.4%; IEA 2035 range 700–1,700 TWh | With a wide 2035 range, phased capacity additions that expand as demand is confirmed are preferable |

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 IEA, Energy and AI (Apr 2025), executive summary
Note: Figures in the data column come from pages 1–4

### Risk matrix: cycle peak and outlook uncertainty matter most [*1,*2,*3]




Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Bank of Korea ECOS market rates (daily) *3 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Anchors: IEA 2035 range ±42% vs base · aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x · KTB 10y 3.81→4.21% · UST 10y 3.79→5.09% (2023-01-02→2026-10-16) · supply chain is qualitative. Positions reflect the author's qualitative judgement

### Next: Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers

#### An earnings-led sector, yet 40x P/E: how should an IPO be priced?

- Vol.2 data-centre PF — hypothetical Korea 40 MW (KRW) and U.S. 100 MW (USD): debt capacity, DSCR and expansion option from a lender's view
- Vol.3 IPO pricing — hypothetical Target Co., assumptions disclosed with their rationale
- Vol.3 overseas peers — Korean peer P/E and EV/EBITDA vs overseas peers (SEC EDGAR)
- Fair value → IPO discount → indicative price range, Monte Carlo sensitivity

Note: Timing and method of Vol.2 and Vol.3 may change

### Limitations of this note

- Ground data covers commercial sales only; data-centre load may be classified as industrial
- IEA figures are summary values entered manually and checked against the source
- The market window (Jan 2023–Oct 2026) and earnings window (FY2022A–FY2025A) do not match exactly
- P/E uses reported NI without one-off adjustments; thin earnings distort P/E
- The sample is four Korean names; overseas peer comparison comes in Vol.3
- The split and headline P/E cover the 3 names with positive base NI — excluding Doosan Enerbility (FY2022A net loss)

Note: To be addressed from Vol.2 onward

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# Expected Q&A: answered with numbers

## Questions likely in investment review or due diligence, with page references

### 11 expected questions [*1,*2,*3,*4,*5]

| Question | Answer | Page |
|---|---|---|
| Isn't AI power demand overhyped? | IEA: 945 TWh in 2030E (2.3x 2024); AI data centres +50% in 2025 vs world +3%. The 2035 range (700–1,700 TWh) is wide | 1 |
| Isn't Virginia a special case? | It is one state. But commercial sales are +73% since 2015 (U.S. +10%) and the gap is widening (6.8%/yr over 3 years). Industrial classification is a stated limitation | 2, 5 |
| Do earnings support it? | NI of 3 names (Doosan Enerbility excluded: FY2022 net loss) rose 5.5x (FY2022A→FY2025A), explaining 61% of the market-cap gain (16.2x) | 3, 4 |
| Has the market priced it in? | For 3 names (ex. Doosan Enerbility), earnings explain 61% and multiples 39%; P/E rose 13.5x → 40.1x | 4 |
| Why is LS ELECTRIC's price up 3x but its market cap 15x? | A 5:1 split on 2026-04-13. The price move KRW 55,100 → 164,600 (2.99x) compares different share sizes; split-adjusted 2.99×5 = 14.9x, matching the cap multiple (shares 30,000,000→150,000,000) | 4, A2 |
| Which is right: 7.7x combined or 14.6x average? | Both — they answer different questions. Combined is a cap-weighted basket; the average weights names equally. Doosan Enerbility (4.4x) was 72% of the starting cap, pulling the combined figure down. Use combined for sector value, average for per-name re-rating | 4 |
| Why is Doosan Enerbility excluded from the split? | P/E = cap ÷ NI, and FY2022A NI was -7,725 (KRW 100mn), ≤0, so the starting P/E and log change are undefined (FY2025A: 848, turned profitable). At 72% of the starting cap it stays in the 4-name combined multiple (7.7x), but the split and headline P/E use 3 names. Its latest P/E of 503x reflects a thin denominator | 3, 4 |
| Why was LS ELECTRIC multiple-led? | Its cap rose 14.9x while NI rose 3.2x (rank 1 of 3, smallest first), and P/E went 18.3x → 86.1x, so the multiple share is 57%. Causes such as orders or business mix are outside this note's data | 4 |
| P/Es are high — is it a bubble? | Headline: aggregate P/E of 3 names (ex. Doosan Enerbility) 13.5x → 40.1x. P/E rose even as earnings grew; whether it is a bubble depends on earnings durability | 4, 5 |
| What comes next? | Vol.2 data-centre PF (Korea 40 MW, U.S. 100 MW), Vol.3 IPO pricing and overseas peers. Vol.2 sizes AI data-centre PF debt from a lender's view; Vol.3 sets an IPO price range reflecting this note's P/E levels and compares overseas peers | 5 |
| How can this be reproduced? | Full code and data sources are public on GitHub: https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio | A3 |

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 U.S. Energy Information Administration (EIA), Electricity retail sales *3 Korea Exchange (KRX) Open API, KOSPI daily trading *4 FSS DART annual reports (consolidated) *5 Company filing (stock split), author's check
Note: Numbers match the body; shaded = data-treatment questions

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# Data notes: sources, datasets, dates and the market-cap check

## LS ELECTRIC split treatment stated · market cap uses listed shares at each date · overseas peers in Vol.3

### Data sources · datasets · as-of dates

| Source | Dataset | Items | As of | Status |
|---|---|---|---|---|
| DART · FSS | Single-company full financial statements (annual report, consolidated first) | Revenue, OP, NI to parent (4 peers) | 2026-10-20 | Collected |
| KRX · Korea Exchange | KOSPI daily trading data | Close, market cap, listed shares | 2026-10-20 | Collected |
| EIA · U.S. EIA | Electricity retail sales (by state and sector, annual) | Retail sales and price by state/sector | 2026-10-20 | Collected |
| ECOS · Bank of Korea | Market interest rates (daily), KRW exchange rates | KTB 3y/10y, USD/KRW | 2026-10-20 | Collected |
| FRED · St. Louis Fed | 10-Year Treasury Constant Maturity (DGS10), KRW/USD (DEXKOUS) | DGS10, DEXKOUS | 2026-10-20 | Collected |
| IEA reports | Energy and AI (Apr 2025), Key Questions on Energy and AI (Apr 2026) | Data-centre power outlook and growth | 2026-10-21 | Verified vs source |

Note: As-of = data collection date (IEA: date checked against the source). Overseas peer comparison comes in Vol.3

### Market-cap check: listed shares at each date (2023-01-02 · 2026-10-19) *1,*2

| Company | Shares 2023-01-02 | Shares 2026-10-19 | Corporate action | Price x (raw) | Split-adj. price x | Cap x | Cap check |
|---|---|---|---|---|---|---|---|
| HD Hyundai Electric | 36,047,135 | 36,047,135 | No change | 12.98x | 12.98x | 12.98x | 0.00% |
| LS ELECTRIC | 30,000,000 | 150,000,000 | 5:1 split 2026-04-13 (ratio matches) | 2.99x | 14.94x | **14.94x** | 0.00% |
| Hyosung Heavy Industries | 9,324,548 | 9,324,548 | No change | 26.06x | 26.06x | 26.06x | 0.00% |
| Doosan Enerbility | 638,308,033 | 640,561,146 | Shares +0.35% — to be checked in filings (negligible effect) | 4.37x | 4.37x | 4.38x | 0.00% |

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Cap = KRX market cap (close × listed shares at each date). Check = close × shares ÷ market cap − 1. Split-adjusted price x = price x × split ratio. Price comparisons use split-adjusted prices. Corporate actions checked against company filings

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# Methodology: official data, validated and calculated into this report

## Every figure is calculated from official-source data, with sources and dates in the footnotes

**Official sources → validation → calculation → report** [*1]

### 1. Official sources

- DART
- KRX
- EIA
- ECOS
- FRED
- IEA reports (verified)

### 2. Validation

- Cap = close × listed shares
- Stock-split check
- Identifier check
- Units and FX

### 3. Calculation

- Indices, CAGR
- Log decomposition
- $\Delta\ln(\text{cap}) = \Delta\ln(\text{NI}) + \Delta\ln(\text{P/E})$
- Exclude base NI $\leq 0$

### 4. Report

- Conclusion titles
- Korean & English
- Sources and dates

Built with an AI coding tool (Claude Code); question design, scope decisions and number verification by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

# Glossary: units and calculations used in this note

**Units: TWh, KRW 100mn, %, x**

**Glossary**

| Term | Definition |
|---|---|
| TWh | Terawatt-hour, one billion kWh; annual electricity use |
| Base Case | IEA's central scenario; the range is IEA's low–high |
| Commercial sector | EIA retail-sales class; data centres may be commercial or industrial by state |
| Index (2015=100) | Year value ÷ 2015 value × 100 |
| CAGR | Compound annual growth = (end ÷ start)^(1/years) − 1 |
| Market cap | Close × listed shares at that date (KRX MKTCAP) |
| Combined multiple | Sum of caps (end) ÷ sum (start); cap-weighted |
| Simple-average multiple | Average of each name's multiple; equal-weighted |
| Stock split | One share becomes several; cap unchanged, price adjusts → compare split-adjusted prices |
| NI to parent | Net income attributable to owners of the parent (DART consolidated) |
| PER | Market cap ÷ NI to parent |
| Log decomposition | Δln(cap) = Δln(NI) + Δln(P/E), an identity since cap = NI × P/E |
| Earnings / multiple share | Each term ÷ Δln(cap) |
| ECM | Equity capital markets: IPOs, follow-ons |
| Indicative price range | Price range set before book-building |
| PF | Project finance, repaid from project cash flows |

Note: Shaded = core calculation terms in this note

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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

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